

Unitree G1 EDU robot Grasping and Manipulation Controller: Quantitative Evaluation Report

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Introduction

The implemented system allows the execution of grasping and manipulation tasks by controlling the Unitree G1 EDU robot model. The operations are carried out using MuJoCo, a physics simulator.

To assess the performance of the controller, the application was tested on two different benchmark tasks through multiple trials to ensure the robustness and reliability of the evaluation.

Benchmark task description

The two tasks were designed to evaluate complementary features of the system:

- **Tabletop pick-and-drop:** in this scenario, the robot had to move a cylinder and a cube into their corresponding containers. The complication consisted of having the two items stacked one on top of the other, since their geometric shapes made the grasping part relatively simple. This configuration required the robot to perform multiple grasping and moving operations to reach its goal, making it necessary to define a long, structured sequence of steps.
- **Object relocation:** in this situation, a mug had to be grasped by its handle and placed on top of a coaster without tipping it over. During this operation, the robot had to be controlled with higher precision and stability, since the space available for the grasping action was significantly smaller.

Both tasks were broken down into multiple parts, each with a specific sub-goal which, when reached, prompted the simulation to move to the next step.

The evaluation encompassed four sets of 20 trials each: a total of 40 trials for each task, half of which were run with fixed object positions, while the other half applied a randomized offset to their spawning locations, with the offset limited to a fixed range. These variations were applied only to the positions of the objects that had to be grasped by the robot.

This testing procedure aimed to emulate the unpredictability of a real-world scenario, where the robot might have to adjust its actions based on small variations in the environment.

Experimental setup

The scope of the assessment was precisely defined through various contextual constraints and parameters to avoid inaccurate results due to variations in the testing conditions.

In particular:

- the **MuJoCo version** was kept unchanged throughout the entire process.
- the tested **robot model** was always the same.
- the same **machine** was used for the evaluation.
- the **simulation parameters** were left unchanged throughout the process:
 - **position error threshold** to grasp the cylinder.
 - **position error threshold** to align the cylinder with the corresponding basket.
 - **position error threshold** to move into position to grasp the cube.
 - **position error threshold** to align the cube with its corresponding basket.
 - **position error threshold** to reach the handle of the mug.
 - **position error threshold** to release the mug.
 - **randomized offset** for the object spawning locations.

Changing even one of these conditions would have made the results of the process inconsistent with one another.

Indeed, after the first evaluation, certain parameters were changed to observe their influence on performance.

Metrics

The evaluation procedure measured different types of information to assess the performance of each trial during the simulation.

1. **Completion time:** the time required to complete a trial of a task (measured in seconds).
2. **Success value:** the level of success achieved in a specific trial. The possible values change depending on the type of task.

For **Tabletop pick-and-drop**:

- **Suc-Suc** means that both the cube and the cylinder have been dropped into their corresponding baskets.
- **Suc-Fail** means that only the cube has been dropped into its corresponding basket.
- **Fail-Suc** means that only the cylinder has been dropped into its corresponding basket.
- **Fail-Fail** means that neither the cube nor the cylinder has been dropped into its corresponding basket.

For **Object relocation**:

- **Success** means that the mug has been placed on the coaster.
- **Failure** means that the mug has not been placed on the coaster.

3. **Maximum pitch** and
4. **Maximum roll:** these metrics provide a way to measure the stability of the robot during the execution of the tasks. The values are absolute and are expressed in radians. In particular, they were measured for the following joints: **left hip**, **left knee**, **left ankle**, **right hip**, **right knee**, **right ankle**, **torso**, and **waist**.
5. **Position error:** this metric is computed only for the final mug placement in the second task, since the first task does not require a precise final position check for either the cube or the cylinder.

Furthermore, the assessment provided statistical data on the performance of each task when considering both fixed and randomized object positions. In particular, for each set of trials, the test computed:

- the **Average completion time**.
- the **Average max pitch** and the **Average max roll** across all the monitored joints.
- For the **Tabletop pick-and-drop** task:

- the **Success-Success rate**.
- the **Success-Failure rate**.
- the **Failure-Success rate**.
- the **Failure-Failure rate**.
- For the **Object relocation** task:
 - the **Success rate**.
 - the **Failure rate**.
 - the **Average position error** of the mug.

Results

The application was tested on both benchmark tasks. Here, the focus is on the statistical information, which provides a more general and balanced evaluation of the execution.

Tabletop pick-and-drop

Tables for both types of evaluation: fixed and randomized spawning locations.

Comparison of the average performance metrics.

Evaluation	Avg. Completion Time (s)	Avg. Max Pitch (rad)	Avg. Max Roll (rad)
Evaluation 1	16.9122	0.3540	0.0239
Evaluation 2	16.9156	0.3520	0.0239

Transition rates for the two evaluation types.

Evaluation	Success–Success	Success–Failure	Failure–Success	Failure–Failure
Evaluation 1	95.00%	5.00%	0.00%	0.00%
Evaluation 2	95.00%	0.00%	5.00%	0.00%

Object relocation

Tables for both types of evaluation: fixed and randomized spawning locations.

Comparison of the average performance metrics.

Evaluation	Avg. Completion Time (s)	Avg. Max Pitch (rad)	Avg. Max Roll (rad)
Evaluation 1	12.6447	0.3535	0.0202
Evaluation 2	12.6115	0.3522	0.0201

Transition rates for the two evaluation types.

Evaluation	Success	Failure	Average position error (mug)
Evaluation 1	45.00%	55.00%	0.1326
Evaluation 2	50.00%	50.00%	0.1298

Discussion

The results provide valuable information for assessing the quality of the application. First of all, it is clear that the second task is much more difficult to complete than the first one. The need for high precision and delicacy when manipulating the mug handle makes the process very challenging. This is reflected in the generally high failure rates observed in the second scenario. On the other hand, the first task requires more time to complete, since it involves manipulating multiple objects.

The stability values are fairly similar between the two tasks. The first task registers a slightly higher average roll, probably due to the greater number of actions required to complete it.

To improve the simulation performance in the second scenario, the **position error threshold** used when reaching the mug handle was reduced from 0.015 to 0.01 meters, making the condition stricter. Indeed, the results were quite promising, especially for the first evaluation type, which achieved a perfect success rate. The results for the second evaluation type also improved and remained acceptable.

Object relocation revisited

Evaluation	Avg. Completion Time (s)	Avg. Max Pitch (rad)	Avg. Max Roll (rad)
Evaluation 1	12.3310	0.3539	0.0208
Evaluation 2	12.5749	0.3540	0.0201

Evaluation	Success	Failure	Average position error (mug)
Evaluation 1	100.00%	0.00%	0.0598
Evaluation 2	55.00%	45.00%	0.1294

The adjustment not only increased the overall **Success** rate, but also reduced the time required to complete the task, showing that the additional time supposedly spent verifying the stricter condition was recovered during the execution of the subsequent steps.

The change also positively influenced the **average position error** of the mug in the evaluation with fixed object positions. However, it produced only marginal improvements when considering randomized object positions.

Final remarks

The controller successfully completed the first benchmark task with a high success rate in both types of evaluation. Randomized object positions slightly increased the completion time, but the overall performance remained robust.

The evaluation of the second scenario yielded a higher failure rate. However, it was possible to significantly improve the performance by changing a single parameter. This suggests that further improvements could be achieved through additional modifications.

The joint pitch and roll values showed that the balancing policy maintained a stable posture throughout both manipulation tasks.